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1 PURPOSE

This document defines the User Requirements (URS), which state what the site needs from Steam Sterilizer S-101, and the Functional Requirements (FRS), which state how the system meets each user requirement. It is the baseline for validating Steam Sterilizer S-101 and its control system (hereafter the system) for the intended use below.

FOR URS :

The purpose of this User Requirements Specification (URS) is to define general and business process requirements for the Steam Sterilizer S-101 Thermal Qualification, hereafter referred to as "the system"). This URS provides a baseline for validation and is developed to document Gilead Science's intended use of the computerized system.

FOR FRS :

The purpose of this Functional Requirements Specification (FRS) is to define the system functionality available in Steam Sterilizer S-101 Thermal qualification, hereafter referred to as "the system"), which fulfill Gilead Science's intended use of the system.

2 BACKGROUND

Steam Sterilizer S-101 is a pre-vacuum steam sterilizer. Its sterilization performance affects patient safety, so the thermal performance, record integrity, and release controls are high impact. The system is managed under GxP and medical device quality system expectations (21 CFR Part 11, 21 CFR Part 820 as revised by the QMSR, and ISO 13485:2016).

3 SCOPE

3.1 In scope

- Thermal performance of cycle C-121 (air removal, heat-up, exposure, exhaust, drying) Control system security, electronic records, electronic signatures, and audit trail
- Cycle review, release, and requalification workflow
- Backup, restore, and power-loss behavior
- Interfaces to the site time server, document management system (DMS), and quality system

3.2 Out of scope

- Facility HVAC and steam generation
- Packaging design and product-specific sterility assurance studies
- Biological indicator studies
- Supply chain and raw material inventory

4 SYSTEM OVERVIEW

The system consists of the sterilizer chamber and door, the steam and vacuum systems, a programmable logic controller (PLC), a human machine interface (HMI), the chamber drain temperature probe TT-101, the chamber pressure transmitter PT-101, and a database for cycle records and the audit trail. It connects to the site time server (NTP), the DMS, and the site quality system.

5 DEFINITIONS

Table 1: List of term definitions

Term	Definition
ALCOA+	Data integrity principles: attributable, legible, contemporaneous, original, accurate, plus complete, consistent, enduring, available
Cold spot	The location in the chamber or load that is slowest to reach, or lowest in, temperature
CPP	Critical process parameter
DMS	Document management system
F0	Accumulated lethality expressed as equivalent minutes at 121.1 °C (z = 10 °C)
FRS/URS	Functional / User Requirements Specification
HMI/PLC	Human machine interface / programmable logic controller
IQ/OQ/PQ	Installation, operational and performance qualification

Table 1: List of term definitions

Term	Definition
LC-01	Load configuration 01: the defined arrangement and quantity of trays in the chamber
NTP	Network Time Protocol
RPO/RTO	Recovery point objective (data you can lose) / recovery time objective (time you can be down)
RTD	Resistance temperature detector
WORM	Write-once, read-many storage

6 USER ROLES

Roles	Permission
Operator	Run cycles (only if trained on S-101)
Supervisor	Run cycles and change cycle parameters
QA Reviewer	View records, apply the Reviewed signature
Validation Engineer	Use validation run mode
Validation Manager	Co-sign requalification and recipe approval
Administrator	Manage users, unlock accounts, enable media ports, restore backups
Service Engineer	Enter calibration data, acknowledge preventive maintenance

7 USER/FUNCTIONAL REQUIREMENTS

Rules used in this section: "shall" marks a mandatory requirement; each FRS item describes one behavior; the audit trail is defined once (FRS-DIN-02 to DIN-04) and other sections refer to it; physical items with no software behavior say how they are verified in IQ. The URS ID is repeated only in the first row of each group.

7.1 General Requirements

URS ID#	General Requirements	FRS ID#	General Requirements
URS-1.1	Supplier Delivers manual, P&IDs, calibration certificates, test records.	N/A	N/A
URS-1.2	The sterilizer shall operate within ambient conditions of 15°C to 30 °C and 20 % to 80% relative humidity	FRS-1.2.1	The control cabinet and HMI shall be specified by the supplier for operation from 15 °C to 30 °C.
		FRS-1.2.2	The humidity shall remain in between 20% to 80%
URS-1.3	The sterilizer's external dimensions, with doors closed, shall not exceed 2.5m wide, 2.0 m deep, and 2.5m high.	FRS-1.3.1	The system door shall close automatically with not exceeding 2.5m wide, 2.0 m deep, and 2.5m high.
URS-1.4	The installed sterilizer shall leave at least 0.8m of clear service access on every side that requires maintenance.	N/A	N/A

URS ID#	General Requirements	FRS ID#	General Requirements
URS-1.5	The sterilizer's operating weight, including a full load and chamber water, shall not exceed 2400 kg.	N/A	N/A
URS-1.6	The sterilizer's floor loading shall not exceed the site floor rating of 500kg per square meter	N/A	N/A

7.2 Business Workflow Requirements

URS ID#	Business Workflow Requirements	FRS ID#	Business Workflow Requirements
URS-2.1	A qualified person reviews and signs each cycle record before load release.	FRS-2.1.1	The system shall restrict the electronic signatures to all users except specific assigned users.
		FRS-2.1.2	The system shall request unique user ID and password combinations. The signature should automatically capture the username, time stamp and position.
		FRS-2.1.3	The system shall maintain the cycle records as a pending review upon every cycle completion and wait for the electronic signature to give the output status.
		FRS-2.1.4	The system shall automatically record the generation of the cycle record, the signature event and any status changes to non-editable audit trail database.

URS ID#	Business Workflow Requirements	FRS ID#	Business Workflow Requirements
URS-2.2	Requalification at a defined interval and after significant change, under change control	FRS-2.2.1	The system shall maintain an immutable, configurable countdown timer/date field for the requalification due date and gives the 30 days warning.
		FRS-2.2.2	The programmable logic controller (PLC) shall continuously monitor critical process parameters (CPPs) such as temperature, time, vacuum depth. If it finds any difference, then the system should change the status from qualified to pending requalification.
		FRS-2.2.3	The system switches to “Pending requalification”, it shall ask user to enter the change control reference number and reason for change before allowing further system testing, otherwise it should display execution blocked sterilizer requires validation/requalification.
		FRS-2.2.4	The system shall feature a restricted “validation run mode” accessible only to user with special credentials. This mode shall permit the execution of empty-chamber temperature mapping or biological indicator (BI) challenge cycles under the recorded change control ID.
		FRS-2.2.5	To restore the qualified status and resume standard production a dual electronic signature is required by two different position officers. The event shall be written in audit trail including the change of control ID and time stamp.
URS-2.3	Only operators with documented training on sterilizer S-101 shall be authorized to run cycles.	FRS-2.3.1	The human machine interface (HMI) system shall require all users to enter their ID and password or scan the ID barcode prior to accessing any operating panels.

URS ID#	Business Workflow Requirements	FRS ID#	Business Workflow Requirements
		FRS-2.3.2	Upon selecting a sterilization recipe and pressing the "Start cycle" button, the programmable logic comptroller (PCL) shall execute a security verification check. Make sure the user profile is found in active user's profile database.
		FRS-2.3.3	The system shall write an immutable entry to the secure audit trial for every cycle initiation or change attempt. The log must capture the timestamp, user ID with the position.
URS-2.4	Each cycle shall be linked to a load record that identifies the load configuration, operator, and date.	FRS-2.4.1	The HMI shall prevent cycle initiation until the operator populates the "Load configuration" field by selecting an option from pre-defined drop-down menu.
		FRS-2.4.2	Upon triggering the start command, the system shall automatically poll the system clock and the active security session to permanently bind the load ID, operator ID details and system time and time stamp and it shall become read only.
		FRS-2.4.3	The automatically generated cycle batch report and the physical "Release Ticket" printout shall display the load configuration, operation name and date.
URS-2.5	A failed cycle shall generate deviation record in the site quality system, and the load shall not be released until the deviation is closed.	FRS-2.5.1	The PLC shall continuously monitor critical process parameters (CPPs). If any critical alarm is triggered (Eg, temperature or vacuum hold failure), the PLC shall flag the cycle status as "FAILED" in the batch database and the system shall open a new deviation record with unique ID and store the details.

7.3 System Security Requirements

URS ID#	System Security Requirements	FRS ID#	System Security Requirements
URS-3.1	Only authorized users with unique IDs can change cycle parameters.	FRS-3.1.1	The system shall check the user login ID and Password to its database, and will print when the user ID and password match otherwise it will block the print request
		FRS-3.1.2	The system shall record the user ID, time stamp, date and reason for change and ask for 2nd verification as well.
		FRS-3.1.3	The system shall mention the user ID, date and time at the top of the page
URS-3.2	The control system shall reject any password that does not meet the password rules defined in the validation plan (Minimum length, upper case, lower case, number and special character).	FRS-3.2.1	The system shall ask user to enter the password until user enter at least 8-digit password containing upper case, special characters and numbers.
URS-3.3	The control system shall lock an account after 5 failed login attempts and record the event.	FRS -3.3.1	The system allows users to try the user ID and Password 3 times continuously, when it fails it directs the IT support team to bypass the temporary lock
		FRS-3.3.2	The system shall send a verification code in user email after 5 failed attempts about account locked and needs to contact the IT support then they will send another verification code in the user phones.
URS-3.4	The control system shall log out an inactive user after a time defined in the validation plan.	FRS-3.4.1	The system shall log out the user after 5 minutes of inactive users and the screen should turn dark after 3 min of inactive.

URS ID#	System Security Requirements	FRS ID#	System Security Requirements
URS-3.5	The control system shall disable or restrict access to removable media ports (such as USB) to authorized users.	FRS-3.5.1	The system shall not support any kind of media connection to the device. The system shall allow the administrator and lead to use media connection with the valid credentials.

7.4 **Regulatory Requirements**

URS ID#	Regulatory Requirements	FRS ID#	Regulatory Requirements
Audit Trail			
URS-4.1	Instruments calibrated with traceability to a national standard. (ISO 13485: 2016 clause 7.6 via 21 CFR part 820)	FRS-4.1.1	System automatically creates logs such as date time and action.
		FRS-4.1.2	Each log entry should capture user ID, time stamp, date, new and old values or Critical Process Parameters (CPPs).
		FRS-4.1.3	The system shall restrict any kind of modification to unassigned users and the assigned user system shall record all the details in the audit logs.
		FRS-4.1.4	The system shall show the date and expiration date which should be higher or greater than previous one and lower or smaller than the next one, if it fails the system shall block cycle initialization and display the critical fault.
URS-4.2	The system shall produce accurate and complete copies of cycle records in human-readable and electronic form for inspection. (21 CFR part 11 g 11.10(b))	FRS-4.2.1	The system shall utilize a uniform layout template for both on-screen and physical paper printing and the system shall generate the uniform structural header data such as unique Cycle ID, Batch number, recipe Name, operator Name and time.

URS ID#	Regulatory Requirements	FRS ID#	Regulatory Requirements
		FRS-4.2.2	The system shall generate the read only PDF and the PDF should be secured.
		FRS-4.2.3	The electronic copy function shall directly transfer records to pre-defined, secure network repository(Write-once-read-many [WORM] server directory) using secure protocols (HTTPS).
URS-4.3	The sterilization process shall be validated in accordance with ISO 17665 before routine use. (ISO 13485 clause 7.5.6; ISO 17665)	FRS-4.3.1	The system shall maintain an internal software state machine governing machine status: Unvalidated, validation mode and routine production approved. The default state for any new or modified recipe shall be initialized as unvalidated.
		FRS-4.3.2	The system shall provide a dedicated “Validation Cycle Mode” accessible only to users with the “Validation Engineer” role.
		FRS-4.3.3	The system shall require a dual electronic signature sequence from the “validation Manager” and “Quality Assurance” roles. The approval command must automatically write a baseline configuration checksum to the master audit trail.
Electronic Signatures			
URS-4.4	Electronic signatures shall show the signer’s printed name, the date and time and the meaning of the signature and shall be linked to its record.	FRS-4.4.1	The system shall render an electronic overlay window that dynamically captures and holds four distinct manifest fields prior to database commitment (Signer’s Printed Name, Date and time stamp Record ID)

URS ID#	Regulatory Requirements	FRS ID#	Regulatory Requirements
		FRS-4.4.2	The system shall programmatically block any user role from editing, stripping, moving or unlinking a committed signature manifest from its target data record.

7.5 Integration Requirements

URS ID#	Integration Requirements	FRS ID#	Integration Requirements
URS-5.1	The control system shall synchronize its clock with the site time server.	FRS-5.1.1	The control system (PLC and SCADA/HMI) shall communicate with the facility designated master clock via a standard network time protocol (NPT) or Simple Network time Protocol (SNTP) query handshake over the local ethernet network layer and system shall completely block standard users from manually changing the time and date.
		FRS-5.1.2	If an NTP synchronization query occurs while a sterilization cycle is actively running in phase, the PLC shall temporarily cache the time adjustment. The PCL shall delay the clock update until the current batch finished.
URS-5.2	The control system shall export each cycle report as PDF file to the site document management system.	FRS-5.2.1	The system shall programmatically generate the PDF filename using a strict, un-editable metadate string format and system shall export the PDF directly to designated person via secure network and in encrypted format.

URS ID#	Integration Requirements	FRS ID#	Integration Requirements
		FRS-5.2.2	The system shall write an entry to the secure audit trail documenting the exact time stamp of the export, the generated file name, the destination path, and the success/failure token received from the DMS.

7.6 Performance Requirements

URS ID#	Performance Requirements	FRS ID#	Performance Requirements
URS-6.1	The system temperature, uniformity, F0, air removal, leak rate, dryness, failure alarm.	FRS-6.1.1	The system shall continuously compare inputs from all active chambers Resistance Temperature Detector (RTD). If the temperature delta between any two active sensors exceeds 1.0C during the sterilization period, an alert shall be thrown.
		FRS-6.1.2	The PLC shall calculate real-time accumulation of lethality (F) at 1-second intervals using the standard thermodynamic formula.
		FRS-6.1.3	The system shall execute an initial pulsing vacuum sequence to ensure deep air evacuation and system shall feature a dedicated leak test cycle.

URS ID#	Performance Requirements	FRS ID#	Performance Requirements
URS-6.2	The chamber shall accommodate load configuration LC-01 of 6 wrapped trays.	FRS-6.2.1	The chamber interior shall be equipped with a multi-tier loading rack featuring a minimum of 6 dedicated shelf tracks. The vertical clearance between tiers shall be manufactured to accommodate standard wrapped trays(Dimensions) without interfering with chamber walls.
		FRS-6.2.2	The loading entry port shall be integrated with either an internal automated barcode scanner array or proximity sensor and the system shall immutably lock the string into the header of the cycle batch report.
URS-6.3	Cycle C-121 shall complete from start to end within 90 minutes.	FRS-6.3.1	The system software shall divide the 90-minute limit into optimized maximum phase thresholds: Air removal: 25 mins, Heating: 35 mins, Cooling: 25 mins and Reset:5 mins. System continuously monitors the phase timers and if it exceeds it time then system will trigger the predictive warning on dashboard.
URS-6.4	The control display shall refresh chamber temperature and pressure at least once every 2 seconds.	FRS-6.4.1	The HMI database tags assigned to the Critical process parameters (CPPs) fir temperature and pressure shall be configured with a maximum update scan class of 1 sec.

URS ID#	Performance Requirements	FRS ID#	Performance Requirements
		FRS-6.4.2	The HMIS shall run a background watchdog script that monitors communications with the PLC. If the data refresh link stalls for more than 2 .0 seconds, the HMI numerical display shall immediately replace the values with an error string and in red color.

7.7 Disaster Recovery Requirements

URS ID#	Disaster Recovery Requirements	FRS ID#	Disaster Recovery requirements
URS-7.1	The system should backup cycle records and the audit trail at least once every 24 hours to a separate storage location.	FRS-7.1.1	The system shall feature an automated, non-resettable background service scheduled to execute a backup sequence every 24 hours. The specific minute of execution shall match the system timestamp of the initial setup to balance network load, and each backup package must compile all relational database files.
		FRS-7.1.2	If the backup service fails to connect to the separate storage location the HMI shall trigger a high-priority advisory alarm banner within 2 seconds and the system shall automatically append an immutable entry to active audit trail detailing the backup event.

URS ID#	Disaster Recovery Requirements	FRS ID#	Disaster Recovery requirements
URS-7.2	The system shall be restorable from backup within 24 hours of failure.	FRS-7.2.1	The system shall automatically compile a dedicated system Restoration report immediately following recovery. The system must write a permanent log to the new active audit trail detailing the recovery event, the backup file data utilized, the user ID who executed it and the verification status token.
URS-7.3	On loss of power during a cycle, the sterilizer shall move to a safe state, keep the records already written and make the cycle as failed.	FRS-7.3.1	The system shall be equipped with hardware uninterruptible power suppl (UPS) that automatically senses line power failure and provides a minimum of 10 minutes of backup electricity to the HMI and critical instrumentation loop.
		FRS-7.3.2	Upon the loss of primary utility power all the critical pneumatic and motorized steam inlet valves shall automatically spring return to their closed position, isolating the steam supply. The chamber vent valve shall open or remain safely modulated to prevent uncontrolled pressure build up and the control system shall boot directly into a recovery state and the physical door latched lock.

7.8 Data Integrity Requirements

URS ID#	Data Integrity Requirements	FRS ID#	Data Integrity Requirements
URS-8.1	Record time, temperature, and pressure at least every 10 seconds.	FRS-8.1.1	Every committed data row shall be bound to an immutable timestamp generated by PLC backplane master clock, resolved down to the millisecond and the database engine shall be configured with deadband recording set to 0.0.
URS-8.2	Time-stamped audit trail of parameter changes: who, what, when, old and new value.	FRS-8.2.1	The HMI run time environment shall programmatically intercept any write command to a critical parameter or system variable before committing the modification to the operating registry. The system shall dynamically generate an un-editable audit event row that automatically captures who, what, when, old value and new value.
URS-8.3	Cycle records protected from alternation and retained per site procedure.	FRS-8.3.1	The system database architecture shall enforce strict write-once-read-many (WORM) logic. Once a sterilization cycle switches to completed or aborted status the database should remove all kinds of permission such as edit delete, overwrite to all the users.

URS ID#	Data Integrity Requirements	FRS ID#	Data Integrity Requirements
		FRS-8.3.2	The system shall feature an automated archiving routine that copies cycle records to a secondary, geographically isolated network directory every 24 hours. The system must verify successful data transmission via a file receipt handshake before marking the local copy as eligible for a log term compression.
URS-8.4	Users shall not be able to change the system date or time.	FRS-8.4.1	The system shall feature an immutable for changing a date and time to anyone.

7.9 Non-Functional Requirements

URS ID#	Non-Functional Requirements	FRS ID#	Non-Functional Requirements
URS-9.1	Door shall not open while the chamber is pressurized.	FRS-9.1.1	The door assembly shall be equipped with a mechanical locking pin or latching ring that is physically driven into a locked position by internal chamber pressure. The mechanical geometry shall physically prevent the door from unlatching if internal chamber pressure exceeds 0.05 bar (50 mbar), operating independently of any electronic control system.

URS ID#	Non-Functional Requirements	FRS ID#	Non-Functional Requirements
URS-9.2	Alram messages shall state the cause in plain English and be shown with an alarm ID.	FRS-9.2.1	The look-up text array shall utilize user-accessible terminology describing the physical event. The system is programmatically prohibited from displaying raw PLC register codes, hex-mapped memory address to end users.
URS-9.3	The sterilizer shall be available for at least 95% of scheduled operating time over 12 months, excluding planned maintenance.	FRS-9.3.1	The database engine shall store the daily complied operating state durations in an immutable historical table for a rolling 2-months period (365 days), preventing or manual alternation by any user profile.
URS-9.4	The supplier shall provide a preventive maintenance schedule and procedures.	FRS-9.4.1	The system shall restrict the closure, acknowledgment, and reset of any PM counter or timer to users assigned the service engineer or system administrator roles.

8 REFERENCES

Reference	Used for
ISO 17665 (moist heat sterilization; check current edition)	Process requirements and validation approach
EN 285 (large steam sterilizers; check current edition)	Leak rate, steam quality, performance values
21 CFR Part 11	Electronic records and signatures
21 CFR Part 820 (QMSR, effective February 2, 2026) incorporating ISO 13485:2016	Quality system, calibration, process validation
ISPE GAMP 5 (Second Edition)	Risk-based approach to computerized systems
FDA guidance on data integrity	ALCOA+ principles
Validation Plan for S-101 (to be written)	Justification of all example values

9. Appendices

Appendix A: Checklist for writing requirements

1. Does the sentence use "shall" and one behavior only?
2. Can you write a test with a clear pass or fail result?
3. Does it contain a number and a unit where one is needed?
4. Does it name a role, not a person?
5. Does it avoid vague words (fast, secure, appropriate, user-friendly)?
6. Is it written once, with other sections referring to it?
7. Does every URS have at least one FRS, or a stated IQ verification?
8. Has the spelling been checked?

Appendix B: Data integrity (ALCOA+) mapping

Principle	Requirements that support it
Attributable	FRS-SEC-01, FRS-DIN-02, FRS-REG-10
Legible	FRS-REG-05, FRS-REG-06
Contemporaneous	FRS-DIN-01, FRS-INT-01, FRS-DIN-07
Original	FRS-DIN-05, FRS-DRC-01
Accurate	FRS-REG-01, FRS-REG-02, FRS-REG-04
Complete	FRS-DIN-01, FRS-DIN-02
Consistent	FRS-INT-01, FRS-REG-05

Principle	Requirements that support it
Enduring	FRS-DIN-06, FRS-DRC-01
Available	FRS-DRC-04, FRS-DRC-05, FRS-REG-06